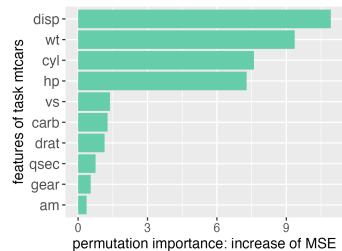
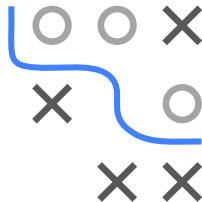


Random Forest Feature Importance



Learning goals

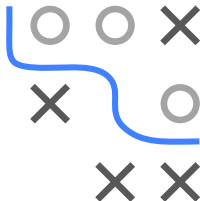
- Understand that the goal of feature importance is to enhance interpretability of RF
- Understand FI based on feature permutation
- Understand FI based on improvement in splits



PERMUTATION FEATURE IMPORTANCE

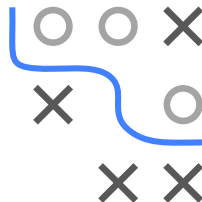
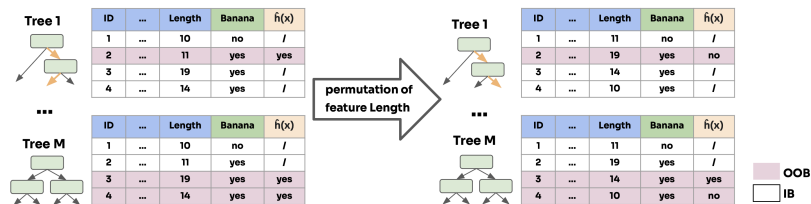
RFs improve accuracy by aggregating multiple decision trees but **lose interpretability** compared to a single tree. **Feature importance** mitigates this problem.

- How much does performance *decrease*, if feature is removed / rendered useless?
- We permute values of considered feature
- Removes association between feature and target, keeps marginal distribution
- Can obtain $\widehat{GE}$ of RF (without and with permuted features) by predicting OOB data, to **efficiently compute FI during training**
- Avoids not only new models (if feature would be removed) but can already use “OOB test data” during training



ID	Color	Form	Origin	Length	Banana
1	yellow	round	domestic	10	no
2	brown	oblong	imported	11	yes
3	green	oblong	imported	19	yes
4	yellow	oblong	domestic	14	yes

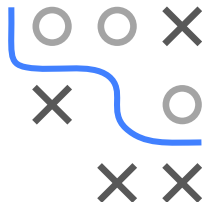
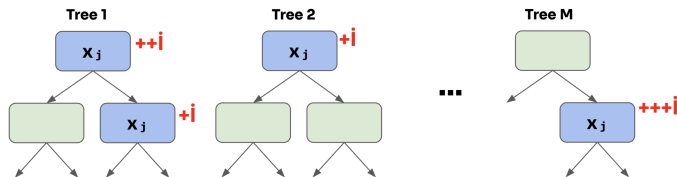
PERMUTATION IMPORTANCE



- 1: Calculate $\widehat{GE}_{OOB}$ using set-based metric ρ
- 2: **for** features $x_j, j = 1 \rightarrow p$ **do**
- 3: **for** Some statistical repetitions **do**
- 4: Distort feature-target relation: permute x_j with ψ_j
- 5: Compute all n OOB-predictions for permuted feature data, obtain all $\hat{f}_{OOB, \psi_j}^{(i)}$
- 6: Arrange predictions in $\hat{F}_{OOB, \psi_j}$; Compute $\widehat{GE}_{OOB, j} = \rho(\mathbf{y}, \hat{F}_{OOB, \psi_j})$
- 7: Estimate importance of j -th feature: $\widehat{FI}_j = \widehat{GE}_{OOB, j} - \widehat{GE}_{OOB}$
- 8: **end for**
- 9: Average obtained $\widehat{FI}_j$ values over reps
- 10: **end for**

IMPURITY IMPORTANCE

Alternative: Add up all *improvements* in splits where feature x_j is used.



-
- 1: **for** features $x_j, j = 1 \rightarrow p$ **do**
 - 2: **for** all models $\hat{b}^{[m]}, m = 1 \rightarrow M$ **do**
 - 3: Find all splits in $\hat{b}^{[m]}$ on x_j
 - 4: Extract improvement / risk reduction for these splits
 - 5: Sum them up
 - 6: **end for**
 - 7: Add up improvements over all trees for FI of x_j
 - 8: **end for**
-

